

CommunitySun:

A Community-centric Approach to Empowering Disenfranchised Populations of Los Angeles

EHS 200D

UCLA Fielding School of Public Health

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June 14, 2024

Executive Summary:

Los Angeles has signed a commitment to be fully transitioned into sustainable energy and carbon-free by 2045, and the city faces significant obstacles to meeting that goal. While not impossible, it will take a concerted effort to make the Los Angeles power grid completely green. Further, there's a noticeable gap in opportunity when recognizing that many low-income communities of color are not granted access to sustainable energy opportunities that more privileged folks are granted access to. One solution to help distribute and make sustainable energy accessible is through the creation of community-based solar projects, where communities invest in a group of solar panels located on buildings in their community or in a local area. Centering equity, there is a focus specifically looking at communities with a high energy burden, or the percentage of income that goes towards paying energy bills. Low-income populations tend to have a higher energy burden due to poor housing conditions or high-energy-consuming appliances, which further drives up their energy usage, putting them into a feedback loop. Energy insecurity can also dangerously lead to service interruptions, which can lead to a plethora of dangerous health outcomes, like heat stress, hypothermia, or cardiovascular disease. In an effort to create a policy solution, an analysis was done to find which census tracts to prioritize in a policy proposal, and an index value was derived to calculate prioritization for certain census tracts based on contributing factors. The policy proposal is a multi-pronged community-centered approach to rolling out community-based solar projects with a lens of energy justice, developed in mind for the California Public Utilities Commission.

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In a time where fossil fuels are rapidly depleting, the implementation of sustainable alternatives becomes increasingly imperative to maintain the demands of our society. The transition from fossil fuels to sustainable resources like solar, wind, hydropower, and others requires a well-faceted approach to ensure a transition in which all needs are met and there is no disruption in the day-to-day lives of millions of individuals. In understanding how to manage greenhouse gases, one must look at the massive contributors. Energy accounts for about $\frac{3}{4}$ of all greenhouse gases emitted ¹. Sustainable energy, as aforementioned, comes in many different forms, and it is now up to larger societies to determine how to integrate it into their society. Contextualizing it down to the city level, some major barriers prevent it from happening, from big utility companies lobbying for their goods to be sold to consumers, geographic disruptions, as well as financial challenges. Understanding solutions that can be created to meet massive cities, where there can be a possible methodology to making sustainable energy more attainable. In comes the city of focus: Los Angeles. Located in Southern California and with a population of nearly 5 million people, Los Angeles is an incredibly interesting location for the implementation of sustainable energy due to its many facets. Los Angeles proper has massive urban sprawl, meaning inconsistent amounts of development over a large geographic area, that continues to develop on the outskirts of the city ². This developing pattern becomes a challenge for trying to push sustainable energy options, creates more greenhouse gas pollution, and poses harm to the surrounding natural environment, which is crucial for the city's wildlife. Further, the city is home to many different types of homes, such as single-family homes, multi-unit homes, ranging upwards of 40 units, and other types of homes that can't straightforwardly embrace sustainable energy due to the nature of their home.

Despite Los Angeles's unique challenges, the city has made commitments to transition to sustainable energy. One could argue this arises from the city's dependency on cars, and the overall carbon footprint of the city already being one of the highest in the world. According to the Global Gridded Model of Carbon Footprints (GGMCF), Los Angeles ranks fifth in the world in terms of overall metric tons of

CO₂ emitted, at 196.4 Mt CO₂, and 158th in the world in terms of per capita Mt CO₂ emission rates. This makes them the second most polluted city in the country, behind New York City and in front of Chicago. The population listed in this model puts Los Angeles' population at 13 million people, most likely accounting for the Greater Los Angeles area, which relates to why their per capita rating is significantly low ³. Decreasing the use of fossil fuels leads to lower greenhouse gas emissions, which in turn can improve conditions in general for communities. Burning of fossil fuels worsens air conditions, leading to serious and significant irritation of the respiratory system and sensitive facial orifices as well as long-term chronic respiratory illness and even death ⁴. In simple terms, fossil fuels are a rapidly depleting, poisoning resource that requires a dramatic transition to sustainable, green energy, looking for solutions that are accessible to greater populations.

As a result, Los Angeles has committed to being 100% clean energy by 2035, an offshoot of an “ambitious climate goal” set by the Biden administration ⁵. As part of Los Angeles's Climate Adaptation Plan, the goal date has been updated from April this year to 2045, with more precise details around mitigation, strategies, and action planning (Climate Action Plan, 2024). But, as aforementioned, Los Angeles still has a long way to go in enrolling all members of the city into this plan. The city must account for disparities in geographic location, access, income, information, and more. The need for sustainable energy solutions is dire.

At the state level, California has a long history of enforcing sustainable energy projects and policies, working to make it more accessible to populations across the state. California launched the Single Family Solar Homes (SASH) program in 2007 and the Multifamily Affordable Solar Homes (MASH) program in 2008, which was the creation of specific types of solar incentives for low-income families and those who were residing in affordable housing. The programs have been redesigned and reimaged, with the launch of the Solar on Multifamily Affordable Housing (SOMAH) program in 2015 and the Disadvantaged Community Single-family Solar Homes (DAC-SASH) program in 2018. Additionally, there have been additional green tariff programs and shared renewable energy to increase access. Assembly bills 2316 and 2838 were passed in 2022 to continue revising these programs as well as

to expand these programs to increase access to them for all Californians. Significant progress towards improving accessibility to renewables includes community-based solar programs.⁶ Community-based solar is defined by the Department of Energy as “any solar project or purchasing program, within a geographic area, in which the benefits flow to multiple customers”⁷. Essentially, community solar aims to be a decentralized model of power, working to benefit consumers away from relying on the typical power grid and empowering communities to be knowledgeable and support their own mode of sustainable power. Community solar operates in a cyclic methodology; consumers or a third party install solar panels on non-residential buildings or create a solar farm adjacent to the community, and consumers purchase panels or credits, creating a “stake” in the program and allowing them to buy the energy generated by the panels.⁸ By localizing sustainable energy, the income generated from the purchase of the panels or the energy stays in the local community becoming a long-term investment. Community-based solar projects aim to be an equitable solution to help Los Angeles in their green energy transition, but there must be a critical analysis alongside an equity lens of understanding how disenfranchised populations in Los Angeles can reasonably benefit from the implementation of community-based solar projects, especially across low-income and energy-burdened populations. More importantly, can community-based solar be an energy justice-centered solution?

In analyzing how the sustainable energy transition impacts marginalized communities, it becomes understood that current paradigms of power will most likely be recreated if there is no careful intention to avoid them. Referencing a 2020 Nature article, the authors call out how the focus of the energy transition has remained on implementing sustainable energy; there needs to be more research done about how these policies and projects can mitigate pre-existing disparities. Mentions of a distributional burden, employment loss due to coal and oil industries shuttering, and lack of decision power in the implementation of sustainable energy solutions all point to the recreation of current-day issues that have arisen with environmental justice⁹ Here, the term energy justice becomes pertinent, a term used to define the empowerment of communities to engage with and make decisions for their energy access. It’s been noted as a three-pronged approach; a distribution of benefits and burdens of energy for communities,

including all stakeholders in the decision-making process to ensure equitable procedural structure, and understanding and rectifying via restorative justice historical and ongoing inequalities⁹.

Zooming into a city-wide lens, certain discrepancies arise when looking at how communities have dealt with the cost of energy historically. Energy burden refers to the amount of income that a household spends on energy, and a home is considered energy-burdened or energy-insecure when it spends more than 6% of its income on energy costs¹⁰. As seen in [Figure 1](#), Angelenos' median energy burden is 2.2%, and the median low-income energy burden is 6%, while comparatively the national rates are 3.1% and 8.1,% respectively. 9% of Los Angeles households have a severe energy burden, classified as above 10%, with additional racial disparities. Over 100,000 Black households and 310,000 Latinx households in the Los Angeles area experience a high energy burden¹¹, depicted in [Figure 2](#). The typical pattern seems to be that households that are earning less tend to spend more of their income on energy, showing a higher burden on these homes. This is coupled with the fact that there's evidence that poorer populations tend to live in older homes or affordable housing that hasn't been properly weatherized, increasing the amount of energy that these families need to consume⁹. Ultimately, this creates a feedback loop of relying on unsustainable energy and also barely being able to afford a good that should be available to them, as shown in [Figure 3](#).

The most troublesome outcome that occurs is service interruptions, as families cannot pay their monthly bills. Families with children who rent their homes are the ones who are most likely to experience these disruptions, and can be especially dangerous during extreme weather temperatures, leading to hypothermia or heat stress due to a lack of adequate utilities. Cold stress can worsen symptoms in folks who already have CVD and those who are at risk of heart attack or stroke, as well as worsen arthritis symptoms, and other respiratory illnesses¹². Heat stress as a result of forgoing cooling systems also has been linked to cardiovascular issues as well as increased sleep disturbances due to the extreme heat, which has been linked to exacerbating mental health conditions¹³. Additionally, other illnesses like cardiovascular, respiratory, and renal illnesses can also experience a worsening in health, ultimately leading to them having to visit hospital emergency rooms as well as overall worse mental health due to

the stress stemming from these shut-offs¹². There's been evidence of folks "coping" or using alternative methods to maintain the temperature of their home, but that can also lead to dangerous results. For example, folks were found using backup generators to power homes, but an incorrect placement can lead to carbon monoxide poisoning for the household¹⁴. Some residents in low-income housing with insufficient heating measures use an unvented gas heater, which has been tracked to have increased levels of nitrogen dioxide and volatile organic compounds in the air as a result of their usage¹⁵. [Figure 4](#) shows how there are many systematic influences on high energy bills, which lead to impacts that can ultimately result in poor health, depicting direct links to understanding how being energy-burdened can lead to adverse health impacts.

In an effort to fix the issue of energy burden, sustainable energy has arisen as a possible solution to improve energy access. Getting one's energy from a source that's not a centralized energy source typically leads to lower costs, as it is localized and not prone to the price fluctuations that come from purchasing energy from a finite source¹⁶. It is also reliable, and as consumers make investments in their solar energy upfront, their month-to-month cost is also driven down a considerable amount.¹⁷ However, there are some critical obstacles to solar implementation that are important to keep in mind when creating policies and projects for these communities. Pre-existing sustainable energy benefits and credit programs are not feasibly appended to low-income marginalized communities. There are many pre-existing barriers: a lack of initial capital to invest in sustainable energy, not a high enough income for a tax credit, renting as opposed to owning their own, a lack of trust, potentially higher energy rates, and a lack of information¹⁸.

Implementing these community-based solutions aims to be a strategic and important method to improving the burden in low-income communities, but understandably, specific communities that bear energy insecurity must be prioritized first, in order to practice the third tenet of energy justice as aforementioned. From here, data analysis was conducted in order to identify which communities must be prioritized in the policy planning process for the implementation of community-based solar programs.

The data analysis we conducted was to identify areas that would benefit from the implementation of a community-based solar program. Using the database Neighborhood Data for Social Change, data about energy burden, median household income, racial identities, and percentage of multi-unit family housing vs single-unit family housing was pulled and created. The aim was to map and understand the possible locations for centering a community-based solar program and cross-reference with the LA Community Solar Opportunities Map that's been created and developed by researchers at the California Center for Sustainable Communities at UCLA¹⁹. Data was collected at the census tract level, and heat maps were prepared to allow for tracking any visual patterns. Energy burden data was divided into two main points: the first was the average percentage of income that households are spending on energy costs and the second was the average percentage of income that low-income households in a tract are spending on energy costs. Using Python and a variety of data libraries, the average energy burden of households in each census tract was mapped, as shown in [Figure 5](#). At first glance, we see West LA, mid-city LA, alongside downtown LA having significantly low energy burdens, hovering around the 1%-3% mark. When looking at the income distribution of these areas, West LA and parts of Mid-City tend to have higher income, allowing for better investment into sustainable energy and technology as a result. Pushing into South LA and into North LA County, such as near Santa Clarita, the burden becomes significantly higher. In the research about these locations, it was found that these communities are more rural and tend to also be lower income, which accounts for the higher burdens found in these areas.

When focusing on the low-income communities where the burden tends to be the highest, we see somewhat similar patterns of distribution. Tracts with the highest energy burdens for low-income families were once again in Northern LA County and were now also in West LA, near the Malibu and Calabasas area. There's a tract near West Hollywood that is also significantly high, depicting the income disparities that are present in this area. It is worth noting that the data collected may not have been collected for every census tract present in Los Angeles County, hence, where there are blank tracts in the county as shown in [Figure 6](#).

This initial data analysis gives an understanding of both where the energy burden is felt in Los Angeles and also where people are concentrated. The census tracts with the highest energy burdens are very populated with people. This signified that there's another level of analysis to be done. What was decided was to create an index, indicating areas of high need alongside potential areas that most benefit from a community-based solar program. The hope is for policies and initiatives created to prioritize these communities first. A low-income score was calculated by subtracting the median household income of the tract as reported from \$250,000, the reported max value for income in the dataset. That difference would then be classified into 5 bins, with a higher rate relating to a lower median income. The multi-unit score was pulled as the percentage of buildings in a tract that housed multiple units, meaning a higher percentage would allow for a higher score. This score allowed the index to prioritize tracts with higher rates of multi-unit buildings that would not typically qualify for the typical state incentives for installing solar on their home. This percentage was also binned into 5, and scores were calculated accordingly from there. The %PoC score was calculated by taking the percentage of white folks per census tract and subtracting it from 100. These differentials were also classified into bins, meaning a higher score means a higher number of folks of color in the tract, and then once again allowing it to be prioritized. Finally, the energy burden percentage was also incorporated, so that a higher score meant a higher energy burden for the tract. The formula is listed in Figure 7 below.

Figure 7

Tabulation Process for Calculating Final Index Score

$$\text{final index score} = \text{low income score} + \text{multiunit score} + \% \text{ of PoC score} + \text{energy burden \%}$$

The data was collected into a spreadsheet and linked [here](#), but was unfortunately unable to be mapped due to some data issues with the dataset. The top 10 tracts are shown in Figure 8 alongside their respective scores. All the index calculations happened in Python with a combination of data libraries. From this analysis, taking an intersectional analysis to understand the different influential factors, we see

that the ideal census tracts to start a community-based solar project are not necessarily following the same distribution of only looking at communities and tracts with a high energy burden. By incorporating other factors, policymakers are able to prioritize tracts that have been historically disenfranchised, incorporating the third tenet of energy justice as mentioned previously.

This data helps inform the policy recommendations this report aims to create. By understanding the factors that influence energy insecurity, a policy in mind would take all of this into account. Of course, this data only begins to tell the beginning of a story. Proper community engagement means engaging the stakeholders within the community with this conversation and allowing them to participate in and create a possible community solar farm together, if possible. Further, the capacity of each of these tracts to host enough solar panels to entirely power the whole community differs greatly. More dense populations require more power, and the amount of roof space and other infrastructure to install solar panels may not be enough. It is recognized that this data is the first step and also does not spell out the whole picture about what the community needs or currently looks like; data is simply a snapshot of how a community operates. Also, capacity for energy and infrastructure is a major determinant variable of being able to enforce how these communities can include community-based solar. Adaptive capacity is the ability of a community to be able to adapt and adjust to changes, and lower-resourced communities tend to have lower abilities to do so⁹. Trying to engage with a community that does not have adequate planning in place for a transition can end up doing more harm than good. Working together with researchers like those at the Center for Sustainable Communities to do a further roof survey and help to understand what the costs of a project will be for the community and its stakeholders is essential to the long-term success of the project.

With all of the aforementioned in mind, the best recommendation is for the California Public Utilities Commission to create a comprehensive plan to incorporate community-based solar into the Los Angeles population, in conjunction with the Los Angeles Department of Water and Power (LADWP). There must be a two-pronged approach, focusing on reducing energy insecurity for families currently and developing and implementing a fully-fledged program to implement community-based solar programs for

marginalized communities in Los Angeles. Reducing energy insecurity would include plans to allow low-income families to invest in energy-efficient appliances and allow for the weatherization of their homes to happen at little to no cost. The CPUC already has an Energy Savings Assistance Program that aims to help low-income families with the weatherization of their homes; a revised update to that program would be to continue to fund it as well as use it as a precursor for families in communities that have been eyed for implementing a community-solar program²⁰. This allows families to consume less energy overall, leading to their energy burden being reduced as well.

The implementation of community solar in Los Angeles can be given the name of CommunitySun, reflecting the core of the project. As for implementing community solar, the first recommendation is identifying neighborhoods that can benefit from and have been historically marginalized, as described in the data analysis above. After the identification of these communities, it is recommended to create a task force, working with stakeholders in the region, including community-based organizations that work with the community, climate and energy justice organizations, community leaders, and, of course, community members, discussing their experiences with energy and utilities. From there, a task force containing folks from CPUC and organizations like the National Renewable Energy Laboratory will develop feasibility plans for implementing community-based solar in these communities. From there, the energy task force meets with the community to inform and discuss what this would look like in their community. It is recommended to use community solar subscriptions, which allow low-income households to buy into the program with low upfront costs²¹. After a community opts in, there will be a 1-year maximum build process of the solar panels on top of community buildings, low-value land, and other sites that have been both approved by the community and zoned. It is recommended to start with a small sample group of families to buy into the solar program, the initial cost is offset via sponsorship and credits from the CPUC. These families can be chosen via an opt-in process. After the initial sample group (recommended anywhere from 10-20 families) works out, it can be recommended to continuously grow the program to a certain number of households a year, establishing a

waitlist of sorts to eventually work towards a majority of the community able to use the community-based solar program.

Alongside the implementation, it is recommended for a thorough feedback process to allow households to give direct feedback to engineers and the CPUC about what is working for their community-based solar programs and what is not. In the vein of representation, it also makes sense to create frequent stakeholder meetings, where folks who are engaging in the program, from the creators to consumers, can meet together to discuss the progress and assess any developments or inquiries anyone in the process may have. In terms of metrics of success, it is also recommended to keep track of how much individual families are spending on energy, from the buy-in process, through a transitional phase, and ultimately a full reliance on the power from the community-solar process. Households should also be reporting their income, and organizers from CPUC should maintain that costs should not be crossing that 6% mark, even holding them to 3% to encourage an equitable approach. CPUC should also develop a project management team to keep track of these different implementations and be able to report back to the state government, but also discuss with larger entities such as the U.S. Department of Energy or solar panel manufacturers to discuss funding and sponsorship opportunities to lower costs. With this, it is hoped that Los Angeles can divert the thousands of hours of sunlight it receives to empowering (literally and figuratively) disenfranchised communities throughout Los Angeles, with the leadership of the CPUC.

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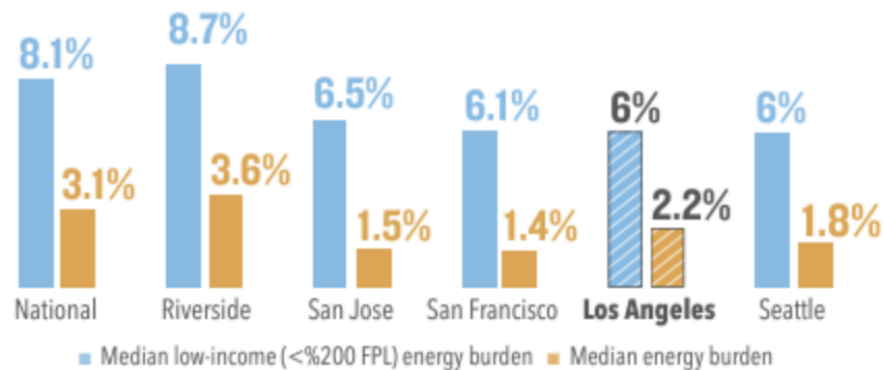
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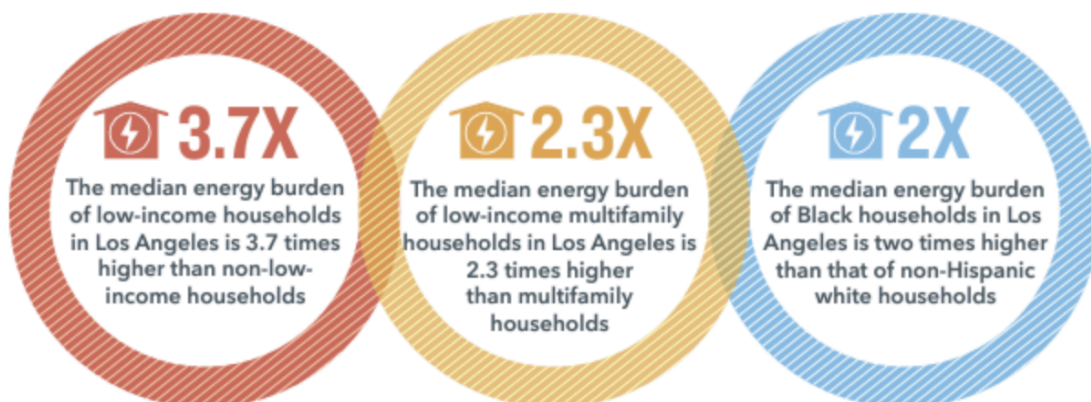
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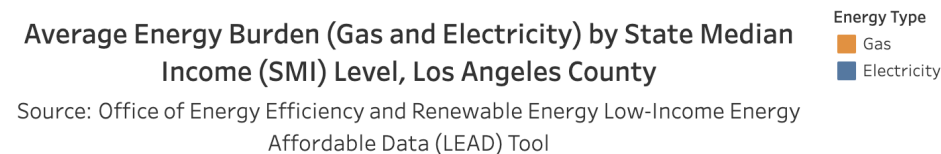
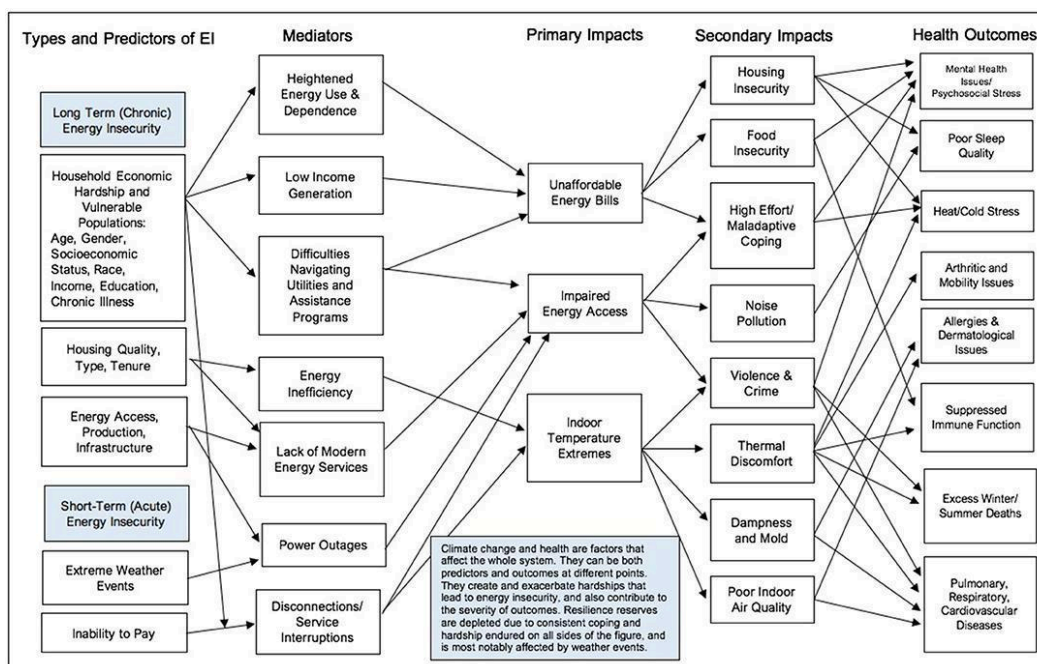
Appendix

Figure 1*Energy Burdens Across the U.S.*

Note: A comparative graph comparing energy burdens to other cities in the U.S.¹¹

Figure 2*Racial Discrepancies Across Energy Burdens*

Note: Source linked¹¹

Figure 3*Average Energy Burden by State Median Income, Los Angeles County**Note: Source¹⁰***Figure 4***Contributing Impacts of Energy Insecurity to Health Outcomes*

Note: Web map describing how different mechanisms lead to worsening health outcomes for already marginalized populations¹²

Figure 5

Map of Los Angeles County and the Median Energy Burden per Census Tract

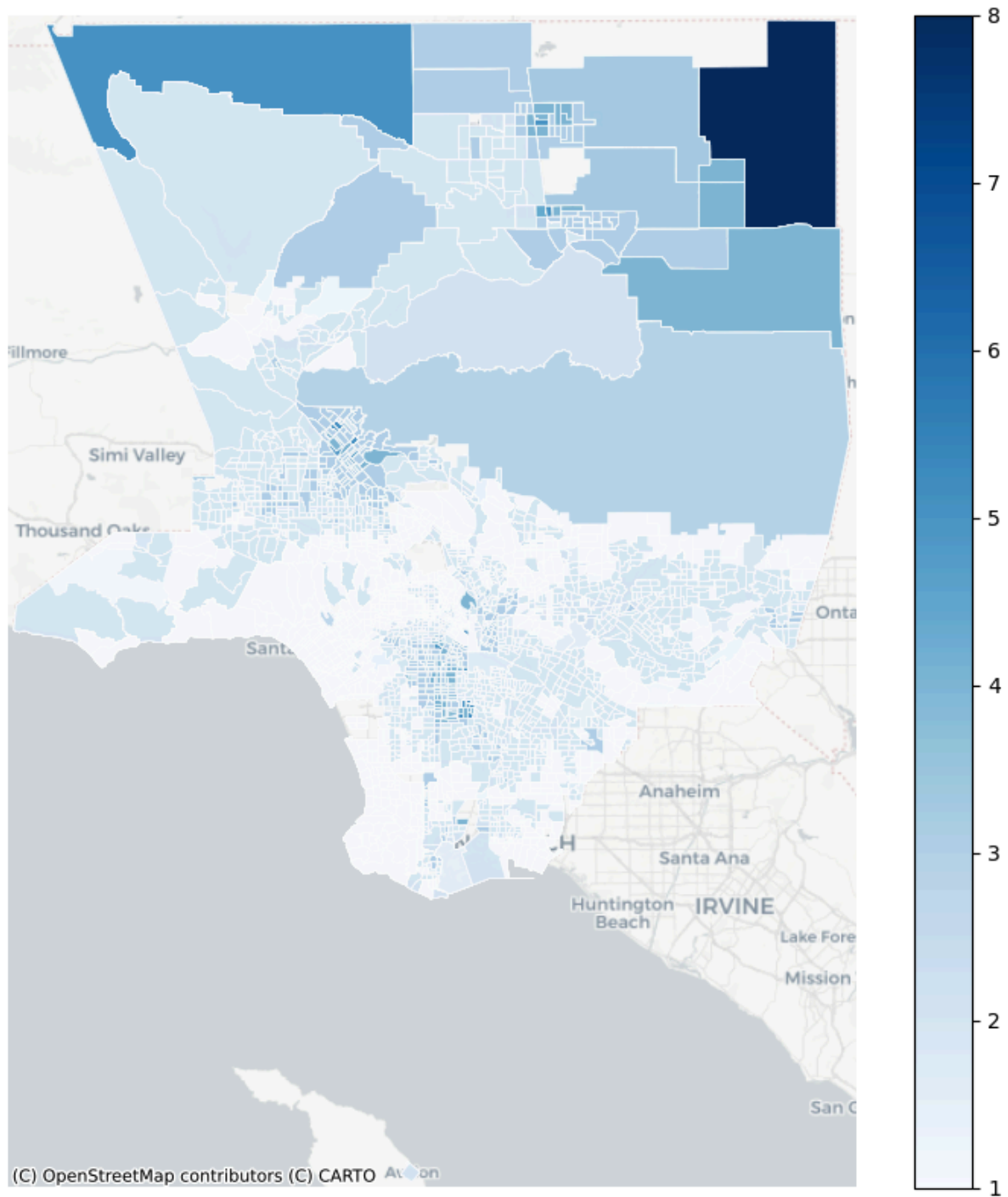


Figure 6

Map of Los Angeles County and the Median Energy Burden per Census Tract for Low-Income Populations

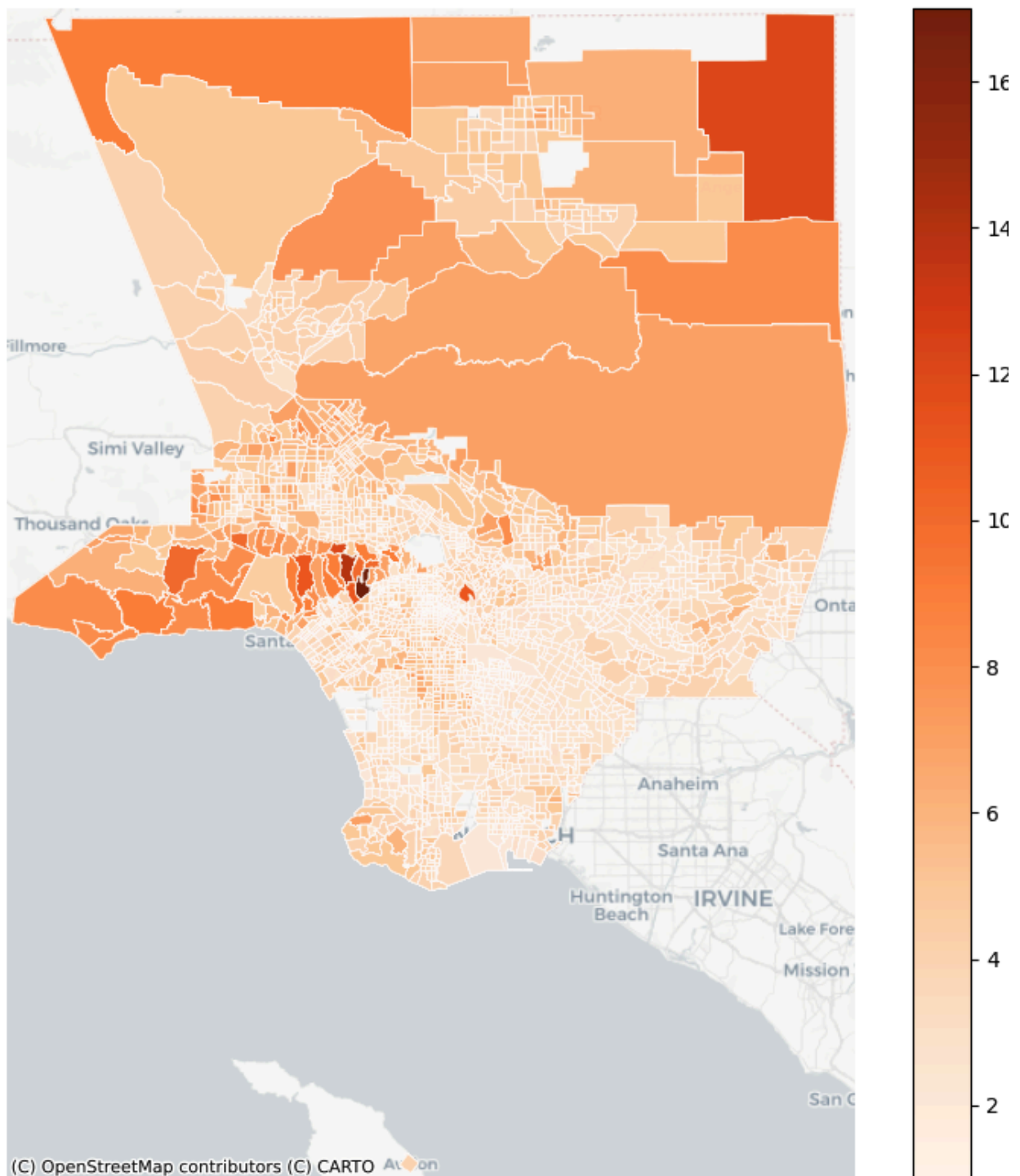


Figure 8*Table of the 10 Census Tracts with the Highest Final Index Scores*

Census Tract ID	Community	Low-Income Score	Multiunit Score	% of PoC Score	Energy burden score (%)	Final Index score
06037242100	South Gate	4	4	4	5.36	17.36
06037910501	Palmdale	4	4	4	5	17
06037104701	Pacoima	4	3	4	5	16
06037206301	Downtown LA/Skid Row	4	3	3	5	16
06037206302	Downtown LA/Skid Row	4	4	3	5	16
06037206303	Downtown LA/Skid Row	4	4	3	5	16
06037231710	Vermont Square	4	4	4	4	16
06037234901	Hyde Park	4	4	4	4	16
06037242600	Compton	4	3	4	5	16
06037571600	North Long Beach	4	4	4	4	16

Note: [Source](#)